

Formative Assessment 3 — Solutions

APM1205: Linear Models

Model and notation. Throughout, the simple linear regression model is

$$Y_i = \alpha + \beta x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

where the x_i are fixed (non-random) constants and the random errors ε_i satisfy the standard Gauss–Markov assumptions:

(L) **Linearity:** $E(\varepsilon_i) = 0$, so $E(Y_i) = \alpha + \beta x_i$.

(CV) **Constant variance (homoscedasticity):** $V(\varepsilon_i) = V(Y_i) = \sigma^2$ for all i .

(I) **Independence:** the Y_i are independent of one another, so $\text{Cov}(Y_i, Y_j) = 0$ for $i \neq j$.

We write $S_{xx} = \sum_i (x_i - \bar{x})^2$ and define

$$m_i = \frac{x_i - \bar{x}}{S_{xx}}.$$

Problem 1 (10 points): Unbiasedness of A and B

(a) $E(B) = \beta$

The least-squares slope is

$$B = \frac{\sum_i (x_i - \bar{x})(Y_i - \bar{Y})}{S_{xx}}.$$

Since $\sum_i (x_i - \bar{x})\bar{Y} = \bar{Y} \sum_i (x_i - \bar{x}) = 0$ (the deviations $x_i - \bar{x}$ sum to zero), the $\bar{Y}$ term drops out of the numerator:

$$B = \frac{\sum_i (x_i - \bar{x})Y_i}{S_{xx}} = \sum_i \underbrace{\left(\frac{x_i - \bar{x}}{S_{xx}} \right)}_{m_i} Y_i = \sum_i m_i Y_i.$$

This expresses B as a linear combination of the observations Y_i , with weights m_i that depend only on the (fixed) x_i 's.

Two properties of the weights m_i will be needed:

$$\sum_i m_i = \frac{\sum_i (x_i - \bar{x})}{S_{xx}} = 0, \quad \sum_i m_i x_i = \sum_i m_i (x_i - \bar{x}) + \bar{x} \sum_i m_i = \frac{\sum_i (x_i - \bar{x})^2}{S_{xx}} + 0 = 1.$$

Now take expectations, using linearity of expectation and the fact that the m_i are fixed constants:

$$E(B) = E\left(\sum_i m_i Y_i\right) = \sum_i m_i E(Y_i).$$

By assumption (L), $E(Y_i) = \alpha + \beta x_i$, so

$$E(B) = \sum_i m_i(\alpha + \beta x_i) = \alpha \sum_i m_i + \beta \sum_i m_i x_i = \alpha(0) + \beta(1) = \beta.$$

Hence B is an unbiased estimator of β . ■

(b) A is linear in the Y_i 's and $E(A) = \alpha$

The least-squares intercept is $A = \bar{Y} - B\bar{x}$. Writing $\bar{Y} = \sum_i \frac{1}{n} Y_i$ and substituting $B = \sum_i m_i Y_i$,

$$A = \sum_i \frac{1}{n} Y_i - \bar{x} \sum_i m_i Y_i = \sum_i \left(\frac{1}{n} - \bar{x} m_i \right) Y_i = \sum_i a_i Y_i, \quad a_i := \frac{1}{n} - \bar{x} m_i.$$

This shows A is also a linear function of the Y_i 's.

As before, two properties of the a_i are useful:

$$\begin{aligned} \sum_i a_i &= \sum_i \left(\frac{1}{n} - \bar{x} m_i \right) = 1 - \bar{x} \sum_i m_i = 1 - 0 = 1, \\ \sum_i a_i x_i &= \sum_i \left(\frac{1}{n} - \bar{x} m_i \right) x_i = \frac{1}{n} \sum_i x_i - \bar{x} \sum_i m_i x_i = \bar{x} - \bar{x}(1) = 0. \end{aligned}$$

Taking expectations,

$$E(A) = \sum_i a_i E(Y_i) = \sum_i a_i(\alpha + \beta x_i) = \alpha \sum_i a_i + \beta \sum_i a_i x_i = \alpha(1) + \beta(0) = \alpha.$$

Hence A is an unbiased estimator of α . ■

Problem 2 (10 points): Sampling variances of A and B

Assumptions used at each step: the linearity assumption (L) was already used above to identify $E(Y_i)$; here the constant-variance assumption (CV), $V(Y_i) = \sigma^2$ for all i , and the independence assumption (I), $\text{Cov}(Y_i, Y_j) = 0$ for $i \neq j$, are what allow the variance of a sum to become a simple sum of variances (no cross-covariance terms).

Variance of B

From Problem 1(a), $B = \sum_i m_i Y_i$ with the m_i fixed constants. For any linear combination of random variables, in general

$$V\left(\sum_i m_i Y_i\right) = \sum_i m_i^2 V(Y_i) + \sum_{i \neq j} m_i m_j \text{Cov}(Y_i, Y_j).$$

By independence (I), every covariance term is zero, leaving

$$V(B) = \sum_i m_i^2 V(Y_i).$$

By constant variance (CV), $V(Y_i) = \sigma^2$ for every i , so σ^2 factors out:

$$V(B) = \sigma^2 \sum_i m_i^2.$$

Now evaluate $\sum_i m_i^2$:

$$\sum_i m_i^2 = \sum_i \frac{(x_i - \bar{x})^2}{S_{xx}^2} = \frac{S_{xx}}{S_{xx}^2} = \frac{1}{S_{xx}}.$$

Therefore

$$V(B) = \frac{\sigma^2}{S_{xx}} = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}.$$

Variance of A

From Problem 1(b), $A = \sum_i a_i Y_i$ with $a_i = \frac{1}{n} - \bar{x} m_i$, again fixed constants. By the same reasoning (independence (I) kills the cross terms, constant variance (CV) lets σ^2 factor out):

$$V(A) = \sum_i a_i^2 V(Y_i) = \sigma^2 \sum_i a_i^2.$$

Expand $\sum_i a_i^2$:

$$\sum_i a_i^2 = \sum_i \left(\frac{1}{n} - \bar{x} m_i\right)^2 = \sum_i \left[\frac{1}{n^2} - \frac{2\bar{x} m_i}{n} + \bar{x}^2 m_i^2\right] = \frac{1}{n} - \frac{2\bar{x}}{n} \sum_i m_i + \bar{x}^2 \sum_i m_i^2.$$

Using $\sum_i m_i = 0$ and $\sum_i m_i^2 = 1/S_{xx}$ (both established above),

$$\sum_i a_i^2 = \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}.$$

Therefore

$$V(A) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) = \sigma^2 \frac{\sum_i x_i^2}{n S_{xx}},$$

where the last equality follows from the algebraic identity $\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} = \frac{\sum_i x_i^2}{n S_{xx}}$ (obtained by writing $S_{xx} = \sum x_i^2 - n\bar{x}^2$ and combining fractions over the common denominator nS_{xx}).

Problem 3 (20 points): Linear transformations of X and Y

Setup. Let B and $SE(B)$ denote the slope and its standard error from the *original* regression of Y on X (Exercise 5.4, Davis's data), and let $t_0 = B/SE(B)$ be the corresponding t -statistic. We use the general fact that for fixed data (x_i, Y_i) ,

$$B = \frac{\sum_i (x_i - \bar{x})(Y_i - \bar{Y})}{S_{xx}}, \quad SE(B) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}}, \quad \hat{\sigma}^2 = \frac{SSE}{n-2}.$$

Numerical values from Exercise 5.4 (Davis's regression)

From the earlier fitted regression of measured weight (Y) on reported weight (X), $n = 101$, $S_{xx} = 4539.3$, $\sum x_i^2 = 329,731$, $SSE = 418.87$, and

$$A = 1.7775, \quad B = 0.97722, \quad \hat{\sigma} = S_E = \sqrt{\frac{418.87}{101-2}} = 2.0569,$$

$$SE(A) = 1.7444, \quad SE(B) = 0.030529, \quad t_0 = \frac{B}{SE(B)} = \frac{0.97722}{0.030529} = 32.01.$$

These numbers are substituted into the general results derived below.

(a) $X' = 10(X - 1)$, Y regressed on X'

Here $x'_i = 10(x_i - 1) = 10x_i - 10$, so $\bar{x}' = 10\bar{x} - 10$ and

$$x'_i - \bar{x}' = 10x_i - 10 - (10\bar{x} - 10) = 10(x_i - \bar{x}).$$

Hence the new sum of squares of X' is

$$S_{x'x'} = \sum_i (x'_i - \bar{x}')^2 = \sum_i [10(x_i - \bar{x})]^2 = 100 \sum_i (x_i - \bar{x})^2 = 100 S_{xx}.$$

Slope. Since Y is unchanged,

$$B' = \frac{\sum_i (x'_i - \bar{x}')(Y_i - \bar{Y})}{S_{x'x'}} = \frac{10 \sum_i (x_i - \bar{x})(Y_i - \bar{Y})}{100 S_{xx}} = \frac{1}{10} \cdot \frac{\sum_i (x_i - \bar{x})(Y_i - \bar{Y})}{S_{xx}} = \frac{B}{10}.$$

Standard error. Rescaling X alone does not change the fitted values $\hat{Y}_i$ or the residuals e_i (the fitted line simply gets re-expressed on the new X' scale), so SSE and hence $\hat{\sigma}$ are unchanged. Then

$$SE(B') = \frac{\hat{\sigma}}{\sqrt{S_{x'x'}}} = \frac{\hat{\sigma}}{\sqrt{100 S_{xx}}} = \frac{1}{10} \cdot \frac{\hat{\sigma}}{\sqrt{S_{xx}}} = \frac{SE(B)}{10}.$$

t -statistic.

$$t'_0 = \frac{B'}{SE(B')} = \frac{B/10}{SE(B)/10} = \frac{B}{SE(B)} = t_0.$$

$SE(B') = \frac{SE(B)}{10}, \quad t'_0 = t_0$

Numerically: with $B = 0.97722$ and $SE(B) = 0.030529$,

$$B' = \frac{0.97722}{10} = 0.097722, \quad SE(B') = \frac{0.030529}{10} = 0.0030529, \quad t'_0 = \frac{0.097722}{0.0030529} = 32.01 = t_0.$$

As required, the slope and its standard error both shrink by a factor of 10 (because $S_{x'x'} = 100 S_{xx}$ enters the denominator of both), while the t -statistic — and hence the conclusion of any test or the significance of the confidence interval for β — is exactly unchanged.

(b) $Y'' = 5(Y + 2)$, Y'' regressed on X

Here $Y_i'' = 5(Y_i + 2) = 5Y_i + 10$, so $\bar{Y}'' = 5\bar{Y} + 10$ and

$$Y_i'' - \bar{Y}'' = 5Y_i + 10 - (5\bar{Y} + 10) = 5(Y_i - \bar{Y}).$$

Slope. Since X is unchanged,

$$B'' = \frac{\sum_i (x_i - \bar{x})(Y_i'' - \bar{Y}'')}{S_{xx}} = \frac{5 \sum_i (x_i - \bar{x})(Y_i - \bar{Y})}{S_{xx}} = 5B.$$

Standard error. Because the transformation is a common linear rescaling applied to every Y_i , the fitted values transform the same way, $\hat{Y}_i'' = 5\hat{Y}_i + 10$, so each residual scales as

$$e_i'' = Y_i'' - \hat{Y}_i'' = (5Y_i + 10) - (5\hat{Y}_i + 10) = 5(Y_i - \hat{Y}_i) = 5e_i.$$

Hence $SSE'' = \sum_i (e_i'')^2 = 25 \sum_i e_i^2 = 25 SSE$, so $\hat{\sigma}''^2 = 25\hat{\sigma}^2$, i.e. $\hat{\sigma}'' = 5\hat{\sigma}$. Since S_{xx} is unchanged,

$$SE(B'') = \frac{\hat{\sigma}''}{\sqrt{S_{xx}}} = \frac{5\hat{\sigma}}{\sqrt{S_{xx}}} = 5 SE(B).$$

t -statistic.

$$t_0'' = \frac{B''}{SE(B'')} = \frac{5B}{5 SE(B)} = \frac{B}{SE(B)} = t_0.$$

$SE(B'') = 5 SE(B), \quad t_0'' = t_0$

Numerically: with $B = 0.97722$ and $SE(B) = 0.030529$,

$$B'' = 5(0.97722) = 4.8861, \quad SE(B'') = 5(0.030529) = 0.152645, \quad t_0'' = \frac{4.8861}{0.152645} = 32.01 = t_0.$$

Here both the slope and its standard error grow by a factor of 5 (matching the rescaling of Y), but again the t -statistic is unchanged from the original regression.

(c) General effect on hypothesis tests and confidence intervals for β

Both parts above illustrate a general principle. Suppose X is rescaled by a factor c_1 (and/or shifted) and Y is rescaled by a factor c_2 (and/or shifted). Then the least-squares slope and its standard error both get multiplied by the *same* factor c_2/c_1 :

$$B_{\text{new}} = \frac{c_2}{c_1} B, \quad SE(B_{\text{new}}) = \frac{c_2}{c_1} SE(B) \quad (\text{for } c_1, c_2 > 0),$$

so that the t -statistic

$$t_0 = \frac{B}{SE(B)}$$

is **exactly the same before and after the transformation**. Consequently:

- The test statistic, p -value, and the conclusion of any hypothesis test $H_0 : \beta = 0$ (or, more generally, of any test on β whose hypothesized value is transformed the same way) are **unaffected** by linear transformations of X and/or Y — statistical significance does not change.
- A confidence interval for β , $B \pm t_{\alpha/2, n-2} SE(B)$, has its endpoints rescaled by the same factor c_2/c_1 as B itself. The interval is therefore still valid and correctly calibrated (same confidence level), but it must be interpreted in the new units of Y_{new} per unit of X_{new} — it is **not** numerically the same interval as before, only proportionally rescaled.

In short: linear (origin/scale) transformations of X and Y change the numerical value of B and $SE(B)$ and the units in which β is expressed, but they leave t_0 , p -values, and the substantive conclusions of tests and confidence intervals for β unchanged (up to the corresponding rescaling of the interval's units).